

Database Management Systems (DBMS)

→ data

unprocessed, unorganised, meaningless

→ Information

- Structured form of data is called as information
- Processed, organised, meaningful, used for decision making

INFORMATION

→ Why we require DBMS?

- easy searching
- Database backup & recovery
- Redundancy (Repetition)
- Security
- concurrency (Accessible across multiple authorised devices)

Name	Roll	Marks
Jur	95	100

→ Database

- Data is a collection of raw facts that have no factual meaning. eg → Jur, 18, 95
- Database is organised collection of related data stored electronically so that it can be easily accessed, managed and updated created by user

→ DBMS

- It's a software that allows user to create, store, retrieve updated and manage it efficiently
- It's an application software

→ Components of DBMS

1. Hardware
2. Software
3. Database Query Language
4. Users
5. Data
6. Procedures

• Hardware is the physical device on which database runs. eg → computer, SSD, server, RAM, network devices.

• Software are the programs that are used to create, manage and access the database. eg → MySQL, ORACLE, NOSQL, DB2.

• Database access language (SQL) is used to communicate with database and is used to create table, retrieve data, update data or access the data.

• Users are those people that interact with a database. Types of users :-

1. Database Administrator, who controls & manage the database.

2. Database Designer, who design the database structure.

3. Application programmer, who develop database application.

4. End users, who use the application to access the data.

• Data is the actual fact & information that's stored in the database.

→ Advantages of DBMS

1. Reduce of Redundancy
 2. Integrity
 3. control data inconsistency
- Redundancy means duplicacy of data, DBMS minimise duplicate copies of data, redundancy means same data is stored multiple times, but in DBMS, the data is stored once & accessed by multiple users.
 - ^{consistency} Integrity means updated data are reflected everywhere for eg → in clg & hostel, same info of students is maintained in 2 diff files and on two different places, suppose if student has changed address in hostel database & forgot to inform college authority so same student's record is diff in two diff files and leads to inconsistency.
 - Integrity means correctness, consistency & completeness of data. It ensures that data value confirm to certain specific tools eg - date can't be like 25/25/2026 ∴ it's invalid, integrity always checks like range check and type of data.

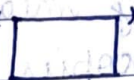
→ Entity Relationship Model (E-R model)

- It's the visual, graphical representation of the database.
- It's a conceptual data model that's used to represent structure of database before implementation. It helps designer what needs to be stored and how different data items are related.
- It's a graphical representation of ~~identity~~ entity, attributes & relationship used during database design phase.
- It's used to understand what user requires.

→ Components of E-R model

1. Entities

- They are real world objects like person, place or anything that can be distinguished from other objects

- Represented using rectangle 

- It's of 2 types tangible & intangible

> Tangible entity

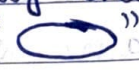
- 1. It's physically present in the world, like book

> Intangible entity

- 1. Virtually present, absent in real world

2. Attribute

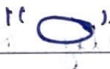
- It's the property / characteristic of entity

- It's represented by oval sign 

> Simple and Composite Attribute

- simple attribute can't be divided further but composite attribute can be divided further
eg → name → last name / first name → composite
roll number → simple

> Single value & Multi value Attribute

- single value attribute has only one value & is represented by single oval 

- Multiple value has multiple values & is represented by double oval 

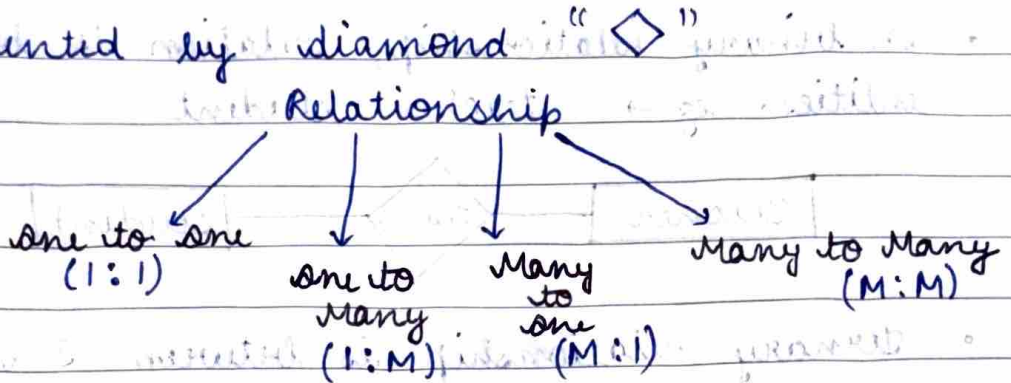
> Stored & Derived Attribute

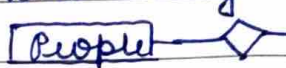
- Derived attribute is found using the already stored attribute and is represented by dotted oval 

- eg → the date of birth is a stored attribute but the age that's derived using DOB is a derived attribute and changes every year.

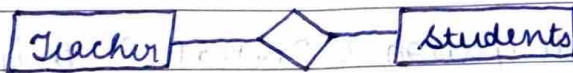
3. Relationship

- It captures how one entity is related to another
- It can relate any no of entities to any no of other entities.
- Represented by diamond "◇"

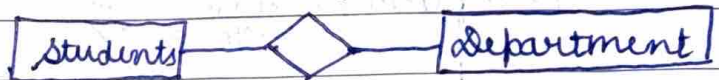


- one to one relationship: one entity can relate to atmost one entity. eg → 
- one to many relationship: one entity can be related to any no of entities

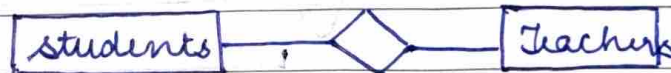
eg →



- Many to one: Many entities related to one entity. eg →



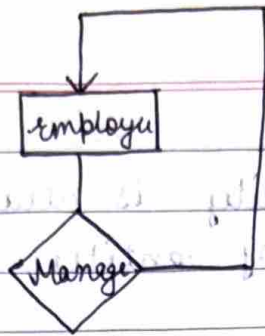
- Many to Many: Many entities are related to many entities. eg →



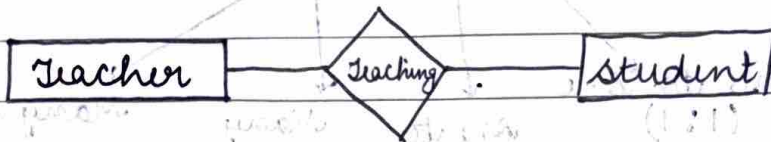
→ Degree of Relationship

1. Unary (Degree - 1)
2. Binary (Degree - 2)
3. Ternary (Degree - 3)

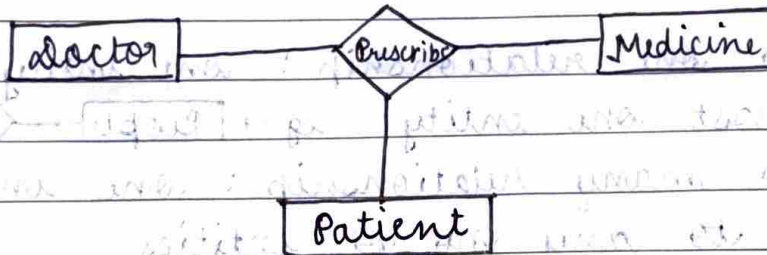
- In unary relationship, the relationship is between one entity and itself. eg → one employee supervise another employee



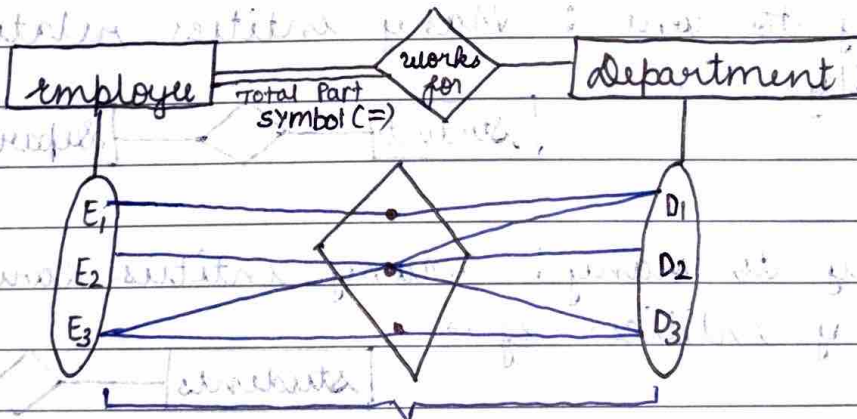
- In binary relationship, relation is between two entities. eg → teacher student



- Ternary relationship is between 3 entities

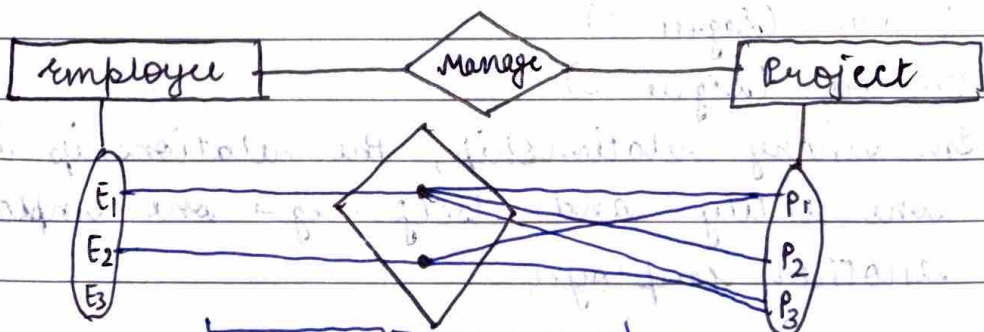


→ Total Participation & Partial Participation



Total Participation

- In total participation, every entity must participate in the relation



Partial Participation

please signed.

- In partial participation, some entities may and may not participate in the relationship. eg → not every employee is a project manager

→ Maximum & Minimum cardinality

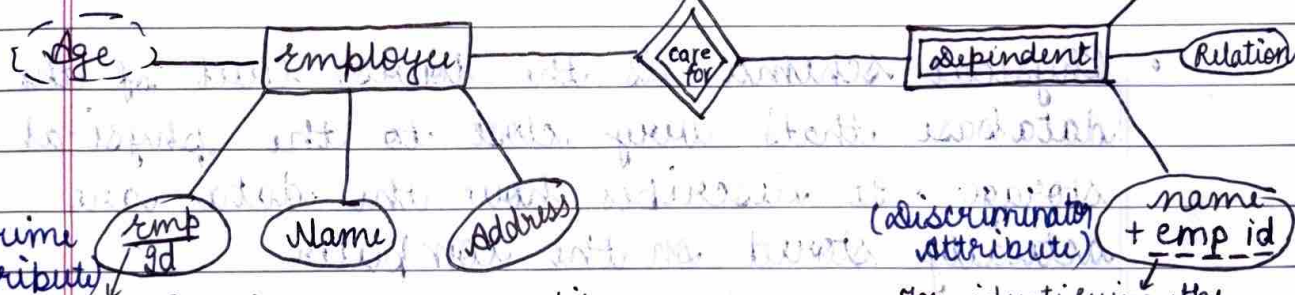
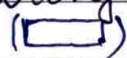
- Maximum cardinality defines maximum number of relationships an entity can have
- Minimum cardinality defines minimum number of times an entity participates in relation

For eg in employee - project ER diagram

Max cardinality = 3

Min cardinality = 0

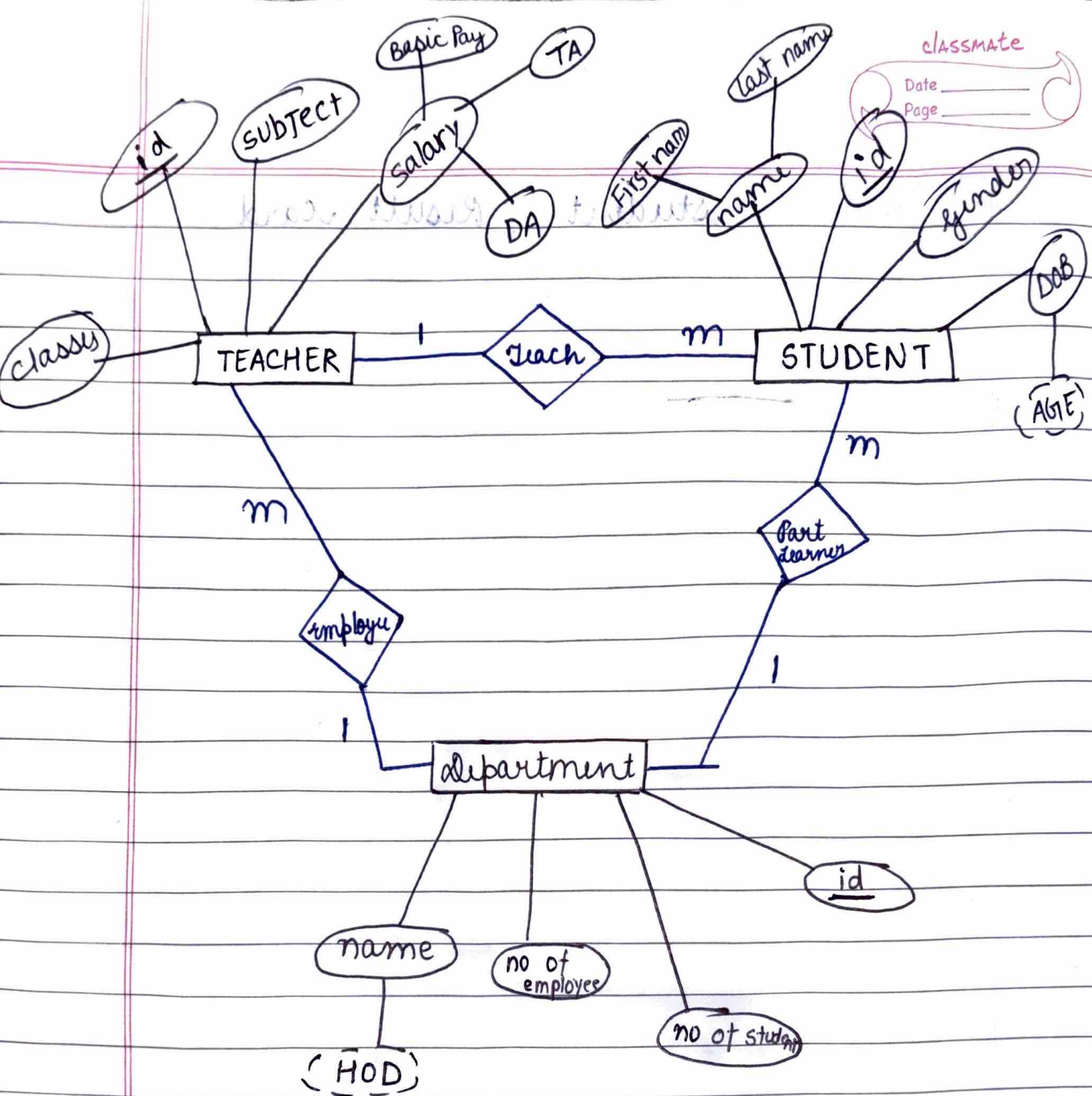
→ Strong & weak entity

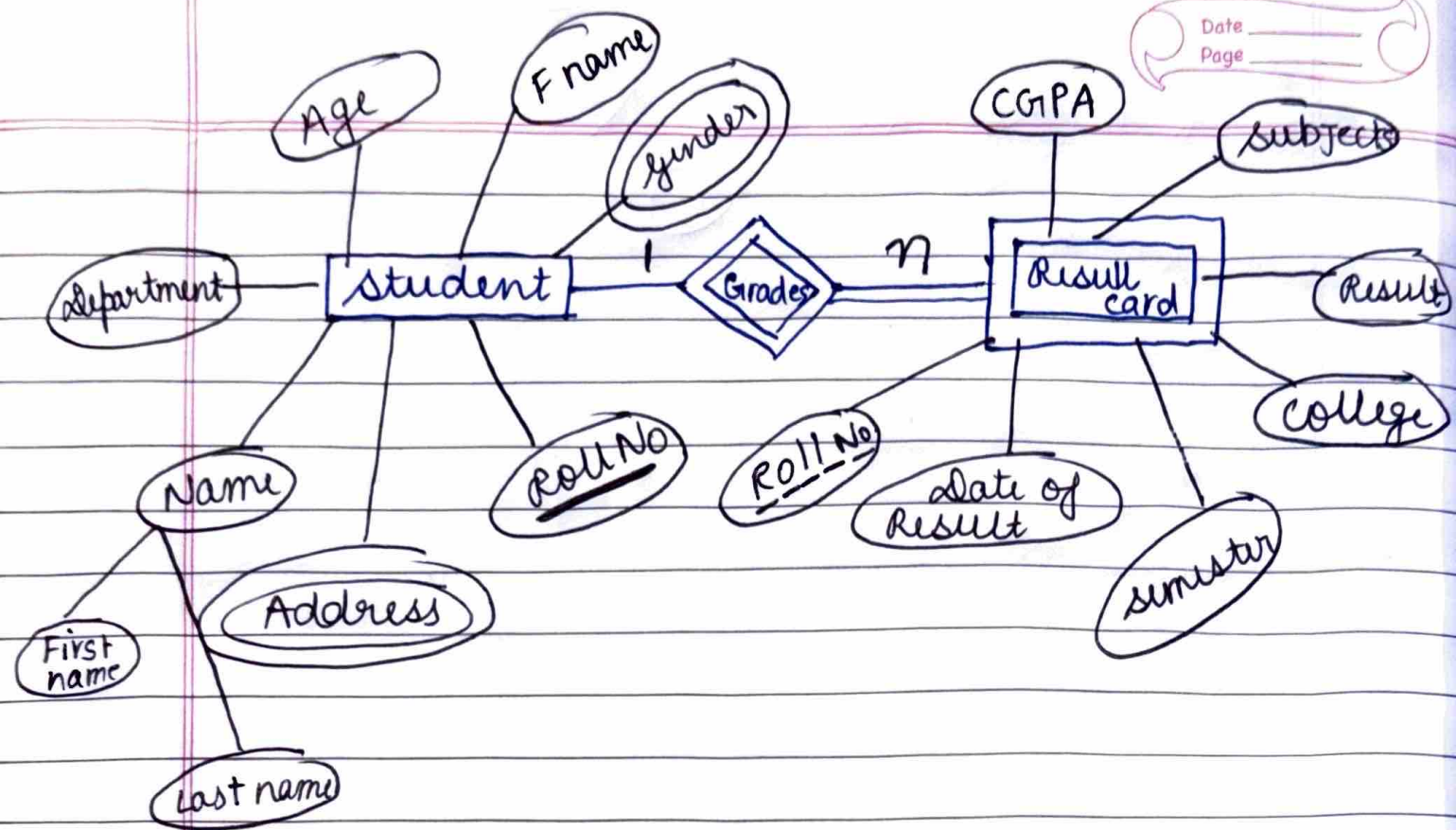


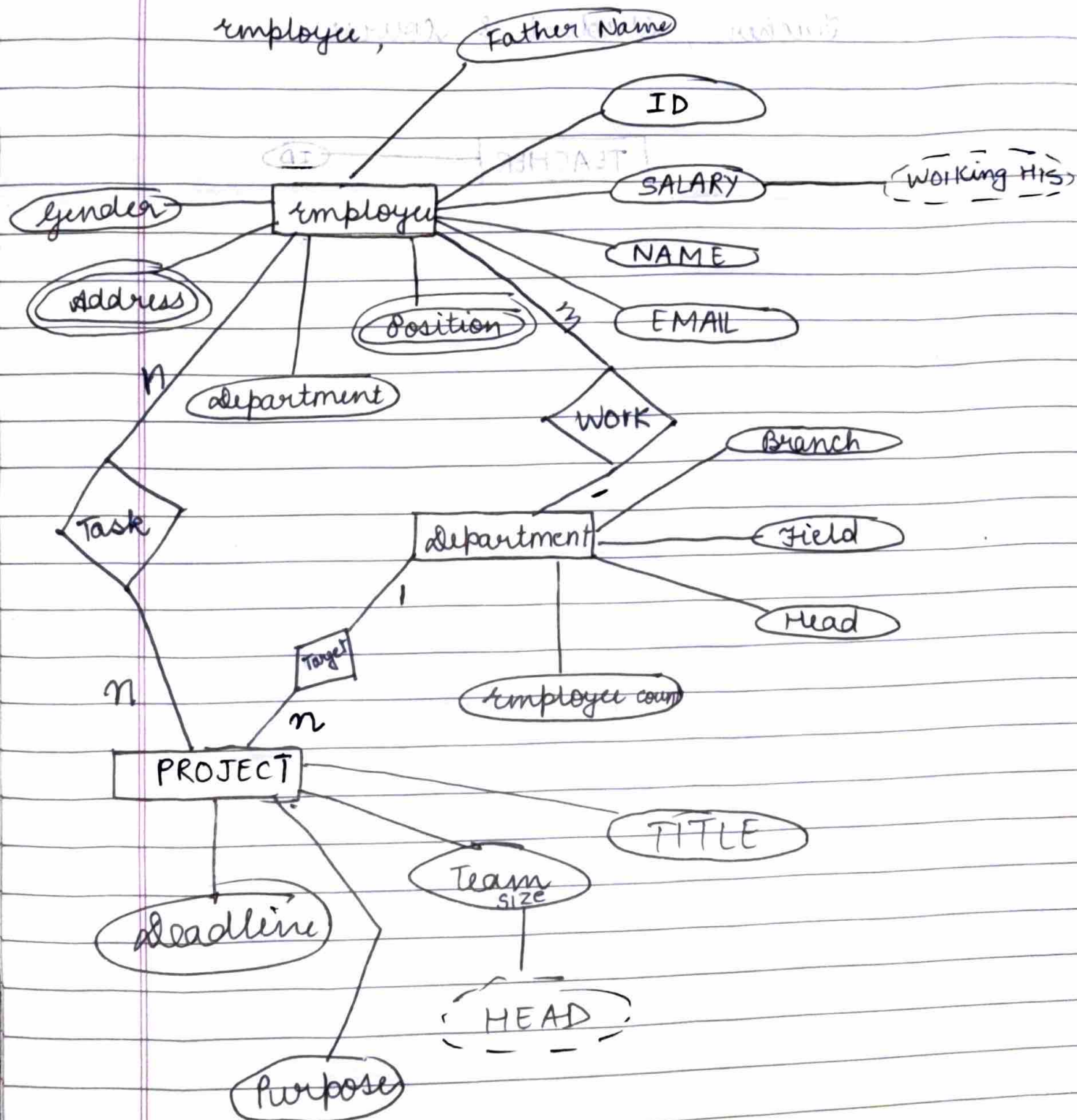
For identifying strong entity

For identifying the weak using primary key of strong

- Strong entity is an entity that has its own primary key and exists independently. eg employee id
- Weak entity doesn't have its own primary key and depends on the strong entity. It can't exist without strong entity. eg → dependents of an employee of a company while seeking medical claims from company



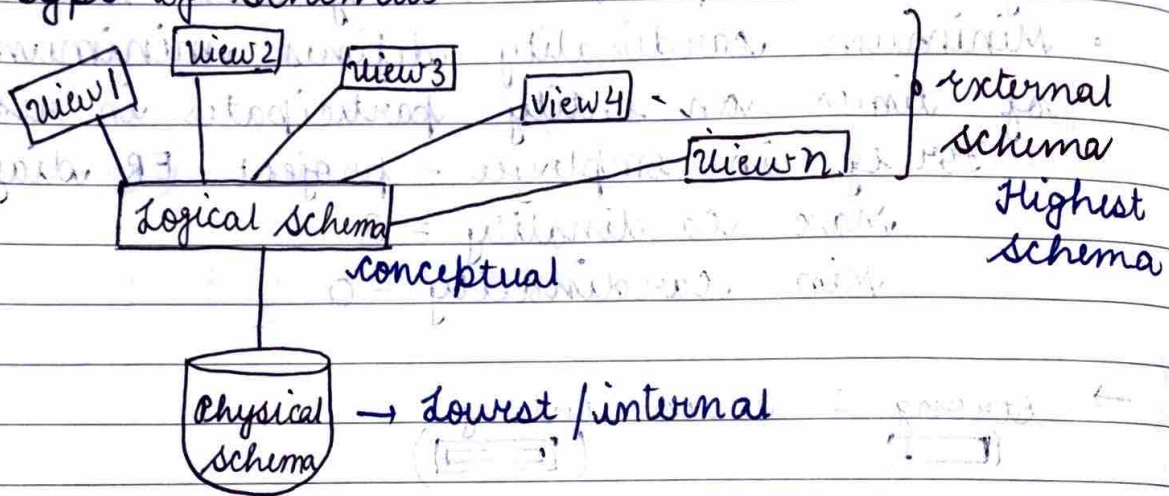




→ Schema ^{changes rarely} designed by database designer

Schema is a logical structure & design of a database that defines how the database is organized, structures like tables, attributes and relationships

→ Type of schemas



- Physical schema is the lowest level of the database that's very close to the physical storage. It describes how the data are actually stored on the computer

- Logical level schema describes what data are actually stored inside database and what's relationship among data

- external level schema is the highest level of database & closest to user. User interact only with related data & aren't concerned with all information contained in the database

- Instance of data or information
- It is the actual content of database
 - It changes very frequently
 - It is updated and deleted by the user

→ Keys in DBMS

student

Name	Address	course ID	
Sonali	C-56	C ₁	→ Rows / tuple / records
Rahul	B-23	C ₂	
Sonali	A-99	C ₁	

- Keys are the constraint / condition / protocol that apply on the value of attribute.
 - These restrictions are specified at the time of defining database through different type of keys
1. Primary Key
 2. Candidate Key
 3. Foreign Key
 4. Alternate Key
 5. Super Key

Course

course ID	Name
C ₁	JAVA
C ₂	PYTHON
C ₃	C++
C ₄	C

- Value of primary key can't be null / repeated
- One table can have only one primary key

→ Primary key

- Primary key of a relational table uniquely identifies every record in the table.
- Value of primary key should be unique & not null.
- Every relation (table) has only one primary key in a database.

→ Candidate key

- The set of all attributes / single attribute that uniquely identify each tuple in a relation are called as candidate key.
- Every table may have one / more candidate key & one of them becomes primary key.
- Minimal super key is called candidate key.

→ Foreign Key

- Used to represent relation between 2 tables.
- A non key attribute of a table which is the primary key of some other table is called foreign key.
- A non key attribute whose value is primary key of some other table.
- Foreign key can be null / be repeated.

→ Alternate Key

- A candidate key that's not a primary key is known as alternate key but alternate key always contain unique value.
- Alternate key may contain null value.

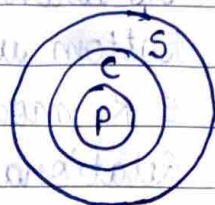
→ Super Key

- Super key is a key that can uniquely identify a record and it may contain more attributes than necessary attributes.

eg → Name & ID

course & email

must have 1 primary key and other variable columns



→ Referential Integrity

- Related to foreign key

student

ID	Name	C-ID	C-ID	Name
101	Rahul	C-109	C-109	C++
102	Amit	C-123	C-100	JAVA

C-101 PYTHON

- It checks all the possible values of foreign key must be subset of all the possible values of primary key

